

COMS3003A Test 1

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This test contributes 5% to your final mark

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1. (5 Points) Provide the equivalent language for the computational problem of:

Given a string $w \in \{0, 1\}^*$, compute the length of the string $|w| \in \mathbb{N}$ in the alphabet $\{0, 1\}$.

- Computational Problem

$$\text{STRING_LENGTH} : \{0, 1\}^* \mapsto \{0, 1\}^*$$

$$\text{STRING_LENGTH}(w) = \langle |w| \rangle_{\mathbb{N}, 2}$$

- Equivalent Decision Problem

$$\text{IS_STRING_LENGTH} : \{0, 1\}^* \mapsto \{0, 1\}$$

$$\text{IS_STRING_LENGTH}(\langle w, \langle n \rangle_{\mathbb{N}, 2} \rangle) = \begin{cases} 1 & \text{if } n = |w| \\ 0 & \text{if } n \neq |w| \end{cases}$$

- ▶ Given a binary string w and an binary represented natural number n , decide if the length of w , $|w|$, is equal to n .

- Equivalent Language

$$L = \{v \in \{0, 1\}^* \mid v = \langle w, \langle |w| \rangle_{\mathbb{N}, 2} \rangle\}$$

- ▶ The set of binary strings that are equal to a string w joined with the binary represented length of w
- ▶ +2 Marks: Correct problem / elements in correct form
- ▶ +1 Mark: is a language $L \subseteq \Sigma^*$
- ▶ +1 Mark: is a binary language $L \subseteq \{0, 1\}^*$
- ▶ +1 Mark: String length is binary natural number

2. (5 Marks) Provide a binary representation for the following data structure:

A graph G is defined by two sets: The set of vertices V , and set of edges $E \subseteq V \times V$

$$G = (V, E)$$

$$V = \{v_1, v_2, v_3, \dots, v_n\}$$

An edge (v_i, v_j) exists between vertices if and only if $(v_i, v_j) \in E$

- +1 Mark: Provided $\langle v_i \rangle$ or $\langle (v_i, v_j) \rangle$
- +2 Marks: $\langle G \rangle$ provided and every edge is accounted for (Need to specify that it is 0 or not in string if cannot be found etc.)
- +2 Mark: Specifies a binary encoding / natively binary representation

For all below answers:

$$e(x) = \begin{cases} 00 & \text{if } x = 0 \\ 01 & \text{if } x = 1 \\ 10 & \text{if } x = \# \\ 11 & \text{if } x = ; \end{cases}$$

Graph

$$\langle v_i \rangle = \langle i \rangle_{\mathbb{N},2}$$

$$\langle (v_i, v_j) \rangle = \langle v_i \rangle \# \langle v_j \rangle$$

$$\langle G \rangle = \langle (v_1, v_1) \rangle; \langle (v_1, v_2) \rangle; \dots; \langle (v_n, v_n) \rangle$$

If $(v_i, v_j) \notin E$ then $\langle v_i, v_j \rangle$ is removed from $\langle G \rangle$

Adjacency matrix (Line breaks included for convenience - not apart of representation)

$$\langle (v_i, v_j) \rangle = \begin{cases} 1 & \text{if } (v_i, v_j) \in E \\ 0 & \text{otherwise} \end{cases}$$

$$\langle G \rangle = \langle (v_1, v_1) \rangle \# \langle (v_1, v_2) \rangle \# \dots \# \langle (v_1, v_n) \rangle;$$

$$\langle (v_2, v_1) \rangle \# \langle (v_2, v_2) \rangle \# \dots \# \langle (v_2, v_n) \rangle;$$

$$\dots;$$

$$\langle (v_n, v_1) \rangle \# \langle (v_n, v_2) \rangle \# \dots \# \langle (v_n, v_n) \rangle$$

Adjacency List

$$\langle v_i \rangle = \langle i \rangle_{\mathbb{N},2}$$

$\forall v_k \in V$, let $E_k = \{v_i \in V \mid (v_k, v_i) \in E\}$ be the list of connected vertices of v_k

$\langle E_k \rangle = \langle v_k \rangle \# \langle v_j \rangle \# \langle v_{j+1} \rangle \# \dots \# \langle v_{j+|E_k|-1} \rangle$ or the # seperated string of connected vertices

$$\langle G \rangle = \langle E_0 \rangle; \langle E_1 \rangle; \langle E_2 \rangle; \dots; \langle E_n \rangle$$